Analysis Sequence "DAAP6 LIPIDS SPOTS NUCLEI MITO MEDIANS bodipy Hoechst cytoplasm"

Modified from previous screen:

+ Changed find cytoplasm channel to hoescht instead of mitotracker because it messed up cell boundaries in PTEN knockouts.

- next time try excluding objects on the side?
- also try modifying the Hoechst channel with find edges etc.

Input Image	Input
	Channel group : 1 **Sequences :** ALL **Flatfield Correction :** Basic Brightfield Correction **Stack Processing :** Maximum Projection

Find Nuclei	Input	Method	Output
	Channel : HOECHST 33342 **ROI :** None	**Method :** B Common Threshold : 0.23 Area : > 43 µm² Splitting Coefficient : 9.4 Individual Threshold : 0.14 Contrast : > 0.11	Output Population : Nuclei

Calculate Morphology Properties	Input	Method	Output
	Population : Nuclei **Region :** Nucleus	**Method :** Standard Area Roundness Perimeter Ratio Width to Length	Property Prefix : Nucleus

Select Population	Input	Method	Output
	Population : Nuclei	**Method :** Filter by Property Nucleus Roundness : > 0.5 Nucleus Ratio Width to Length : > 0.26 Boolean Operations : F1	Output Population : Nuclei Selected

		and F2	

Find Cytoplasm	Input	Method	Output
	Channel : HOECHST 33342 **Nuclei :** Nuclei Selected	**Method :** D Individual Threshold : 0	

Calculate Morphology Properties (3)	Input	Method	Output
	Population : Nuclei Selected **Region :** Cell	**Method :** Standard Area Roundness Ratio Width to Length	Property Prefix : Cell

Find Spots	Input	Method	Output
	Channel : Alexa 488 **ROI :** Nuclei Selected **ROI Region :** Cell	**Method :** B Detection Sensitivity : 0.32 Splitting Sensitivity : 0.811 Calculate Spot Properties	Output Population : Spots

Calculate Intensity Properties	Input	Method	Output
	Channel : MitoTracker Deep Red **Population :** Nuclei Selected **Region :** Cell	**Method :** Standard Mean Sum	Property Prefix : Intensity Cell MitoTracker Deep Red

Calculate Intensity Properties (3)	Input	Method	Output
	Channel : Alexa 488 **Population :** Spots **Region :** Spot	**Method :** Standard Mean	Property Prefix : Intensity Spot Alexa 488

Find Image Region	Input	Method	Output
	Channel : Alexa 488 **ROI :** None	**Method :** Common Threshold Threshold : 0.36 Area : > 0 px²	Output Population : Lipid_droplets Output Region : Lipid_droplets

Calculate Morphology Properties (2)	Input	Method	Output
	Population : Lipid_droplets **Region :** Lipid_droplets	**Method :** Standard Area	Property Prefix : Lipid_droplets

Calculate Intensity Properties (2)	Input	Method	Output
	Channel : Alexa 488 **Population :** Lipid_droplets **Region :** Lipid_droplets	**Method :** Standard Mean	Property Prefix : Intensity Lipid_droplets Alexa 488

Define Results

Results

Method : List of Outputs
Population : Nuclei
Apply to All : None

Population : Nuclei Selected
Number of Objects
Nucleus Area [µm²] : Median
Nucleus Roundness : Median
Cell Area [µm²] : Median
Cell Roundness : Median
Cell Ratio Width to Length : Median
Intensity Cell MitoTracker Deep Red Mean : Mean
Intensity Cell MitoTracker Deep Red Sum : Sum

Population : Spots
Number of Objects
Relative Spot Intensity : Median
Spot Area [px²] : Median

Population : Lipid_droplets
Lipid_droplets Area [µm²] : Sum
Intensity Lipid_droplets Alexa 488 Mean : Mean

Object Results
Population : Nuclei : None
Population : Nuclei Selected : None
Population : Spots : None
Population : Lipid_droplets : None

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